C2 - Contact-event Phase Synchronisation 详细流程

Reference / Event Preparation

保留完整 C1 FIR-QP controller
h, S, AR, q/r/slew, G
pitch-recovery law 不变

从 FDDP reference 读取计划接触转换
left / right touchdown and liftoff
$t_{event,ref}$

Online Phase Alignment + C1 Correction

Toe/heel contact observer
sensor values → site active → leg active
observed events: {leg, kind, t_{obs}}

在时间窗内匹配 observed 与 expected event
$\varepsilon_t = t_{obs} - t_{expected}$
matched / skipped / unmatched

计算有界 phase-offset update
$\Delta t_{target} = clip(t_{expected} - t_{obs}, \pm\Delta t_{max})$
$\Delta t_\varphi{}^{+} = clip(\Delta t_\varphi + clip(K_\varphi(\Delta t_{target} - \Delta t_\varphi), \pm\delta t_{step}), \pm\Delta t_{max})$

查询 phase-aligned reference time
$t_\varphi = clip(t + \Delta t_\varphi, 0, T)$

按相位时间插值名义参考
$q_{nom} = q_{ref}(t_\varphi)$
$\theta_{ref} = \theta_{ref}(t_\varphi)$

运行未改变的 C1 FIR-QP correction
$e_k \rightarrow E_{free} + SA \rightarrow A^* \rightarrow \alpha_k$

形成运行时位置命令
$q_{cmd} = q_{ref}(t_\varphi) + G\alpha_k$

记录 phase / contact / pitch / QP evidence
matched & skipped events / timing error / phase offset
contacts / pitch metrics / correction / solve time

当前证据边界：double-support CoM-shift reference 没有计划接触转换。本阶段支持 contact monitoring、event-capable implementation 与 C1 compatibility，但尚未验证 walking-gait phase synchronisation。